

Day-Ahead Power Price Forecasting (DE-LU & GB) - Executive Summary

Key Takeaways: Across expanding-window walk-forward evaluations, CatBoost is the statistically dominant model in DE-LU with an out-of-sample MAE of 20.58 EUR/MWh (vs Naive-24h 27.98 EUR/MWh, +26.5% improvement, Diebold-Mariano $p=0.0000$). In GB, LightGBM achieved an out-of-sample MAE of 16.18 GBP/MWh (vs Naive-24h 21.67 GBP/MWh, +25.4% improvement, $p=0.0000$). SHAP attribution confirms residual load and kinetic wind generation are the primary price drivers in DE-LU, whereas GB is predominantly governed by gas generation spark spreads and interconnector transfers.

Germany-Luxembourg (DE-LU) Performance

Model	MAE (EUR)	RMSE (EUR)	sMAPE (%)	DM p-val	DM Winner
Naive-24h	27.98	41.48	48.7%	Bench	-
Naive-week	34.81	48.32	54.6%	0.0067	-
Seasonal-naive-avg	29.62	39.94	47.0%	0.1876	-
Climatological-mean	88.39	97.32	74.6%	0.0000	-
LightGBM	21.93	30.16	37.4%	0.0005	YES
XGBoost	21.84	29.35	37.4%	0.0003	YES
CatBoost	20.58	28.19	36.5%	0.0000	YES
LSTM	33.77	45.68	46.1%	0.0019	-
Hybrid	24.04	33.23	40.9%	0.0280	YES

Great Britain (GB) Performance

Model	MAE (GBP)	RMSE (GBP)	sMAPE (%)	DM p-val	DM Winner
Naive-24h	21.67	36.02	33.5%	Bench	-
Naive-week	26.86	43.80	39.5%	0.0090	-
Seasonal-naive-avg	21.60	34.55	30.2%	0.9458	-
Climatological-mean	41.23	49.96	44.2%	0.0000	-
LightGBM	16.18	26.93	23.9%	0.0000	YES
XGBoost	16.56	27.67	24.1%	0.0000	YES
CatBoost	16.46	27.07	24.2%	0.0000	YES
LSTM	25.31	38.69	35.1%	0.0093	-
Hybrid	19.64	31.82	27.5%	0.0530	-